FINAL
PRESENTATION

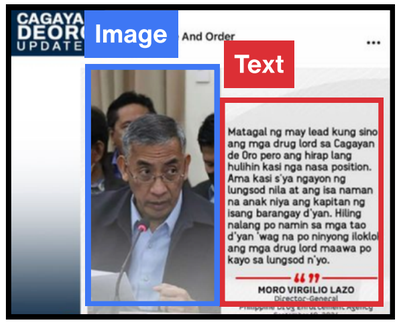
EARLY DETECTION OF MULTIMODAL FAKE NEWS

VIA REINFORCED PROPAGATION PATH GENERATION

L. ZHANG ET AL. · IEEE TKDE 37(2), 2025

Introduction

Time



News
(Fabricated)

Share

Share

Share

User#01

User#07

User#25

Propagation path

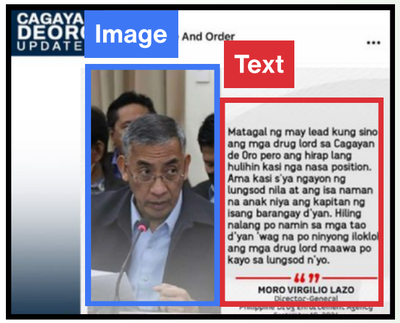
Introduction

Time

Content-based Detection

- Real Time
- Accuracy

Hazard Level: Low



News
(Fabricated)

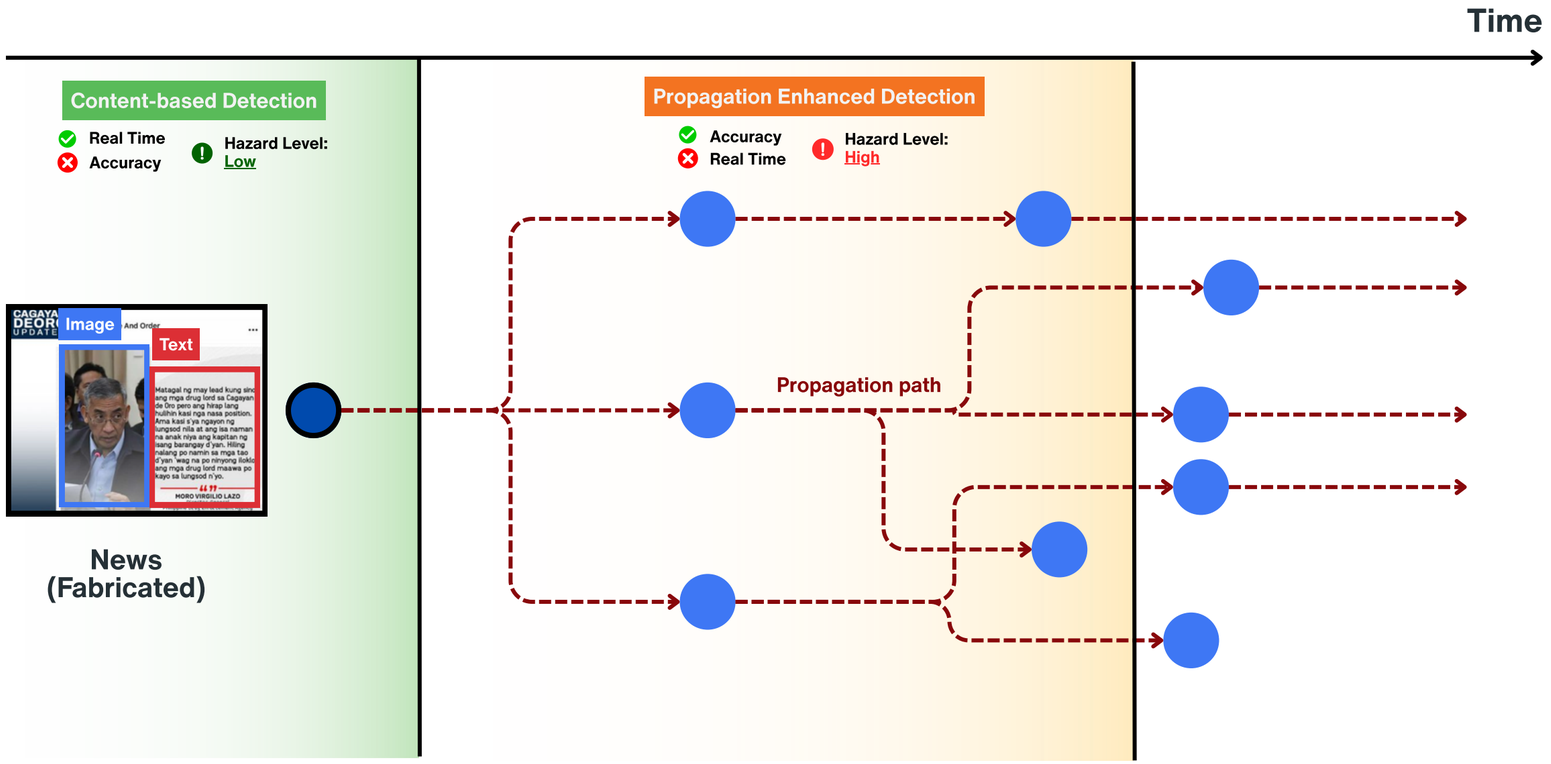
Propagation Enhanced Detection

- Accuracy
- Real Time

Hazard Level: High

Propagation path

Introduction



*Can we leverage the **abundant** social context in the **propagation paths** to enhance the **early-stage detection** before widespread dissemination ?*

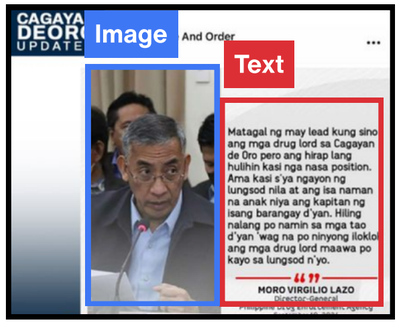
Introduction

Time

Reinforced Propagation Path Generation Fake News Detection (RPPG-Fake)

- Accuracy
- Real Time
- Hazard Level: Low

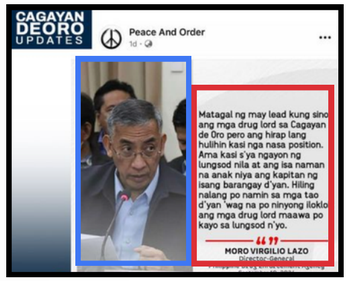
RL Generator



News (Fabricated)

Propagation path

Generate the propagation path (G')

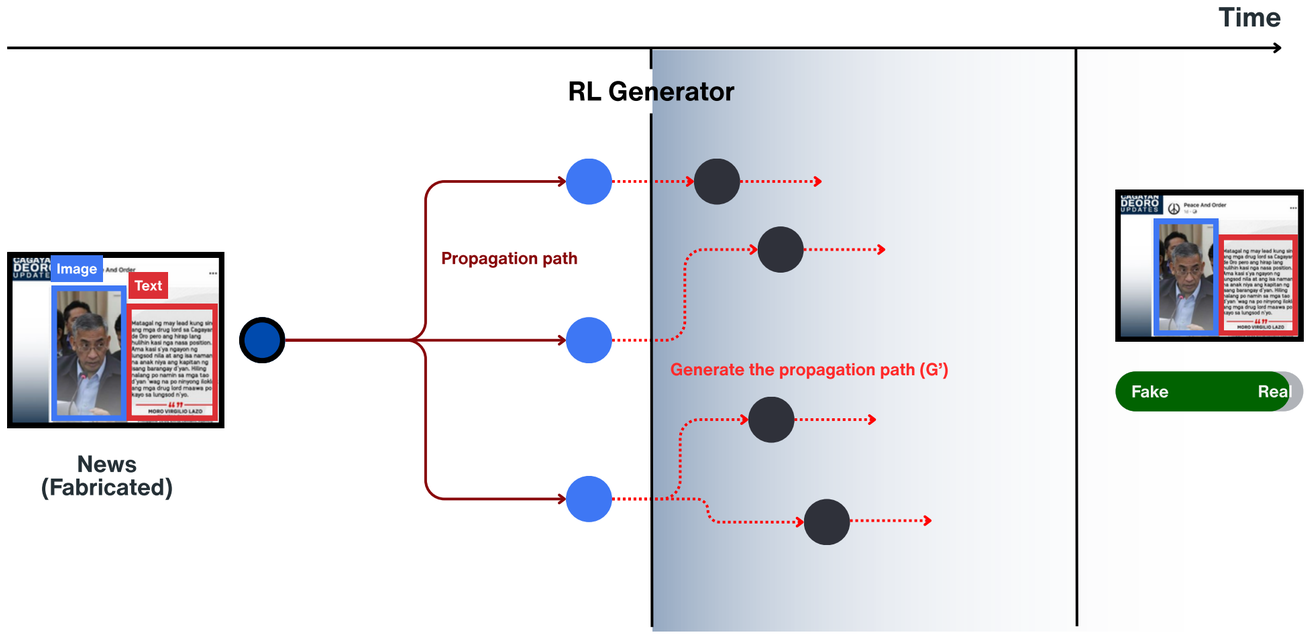


Fake Real

Introduction

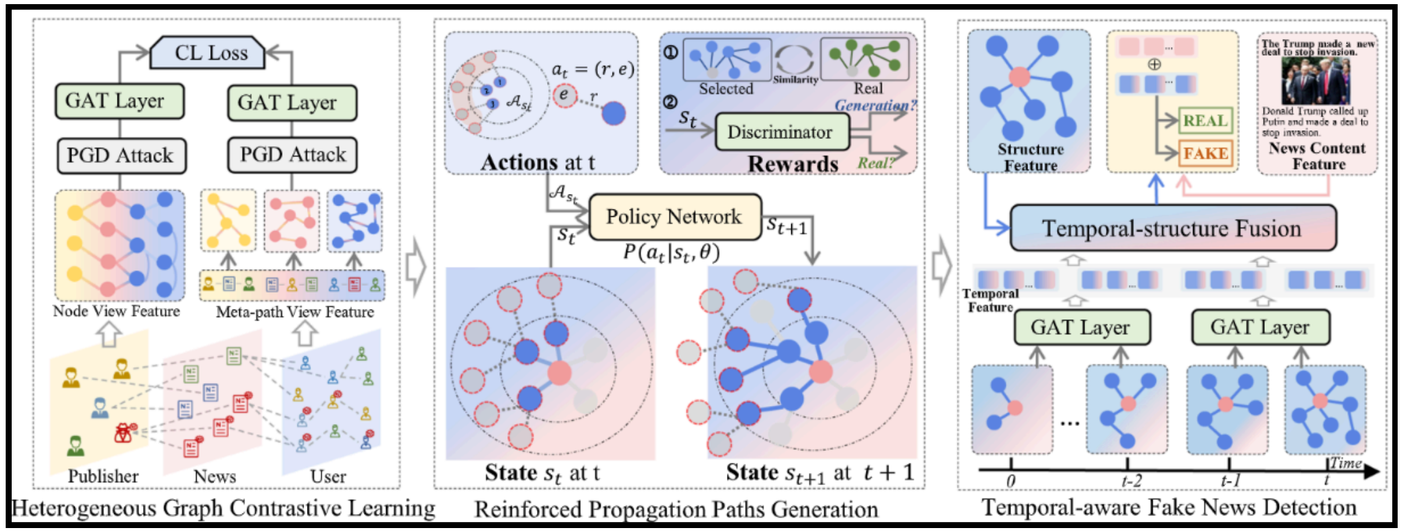
We are the first to explore the challenge of **early fake news detection by generation propagation paths**, which leverage the advantages of extensive social context while mitigating social hazards.

RPPG-Fake model has been proposed. The model can generate the reasonable propagation paths via **reinforcement learning, and furthur integrate propagation tropology structure and temporal depth information through a temporal-aware fake news detection module.**



Methodology

$$P(H \mid P,N,U)\cdot P(G' \mid G,H)\cdot P(y \mid n,G') = P(y \mid n,G)$$



$$P(H \mid P,N,U)$$ → $$P(G' \mid G,H)$$ → $$\cdot P(y \mid n,G') =$$

H: Heterogeneous Graph
P: Publisher
N: News
U: Users

G': Generated Propagation Path
G: Early Propagation Path
H: Heterogeneous Graph

y: News Label ; 1 = Real, 0 = Fake
n: News details (Textual + Visual)
G': Generated Propagation Path

Data

Model training

News

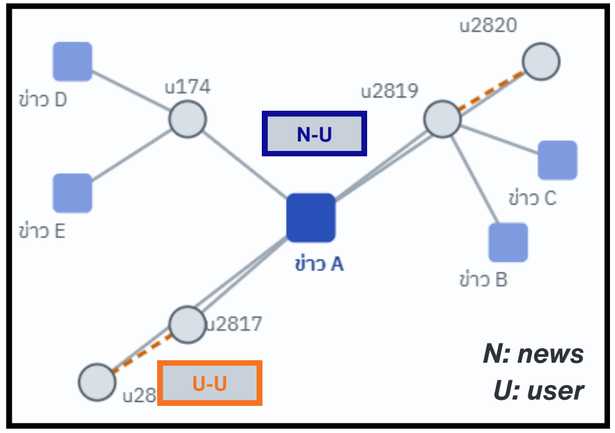
BREAKING: Pope Francis Just Backed Trump, Released Incredible Statement Why-SPREAD THIS EVERYWHERE

Trump is getting support from every leader, and that's the support that will make him grow great and strong. These elections will bring an immense change in our country.

Visual content

Textual content

1 H graph / dataset

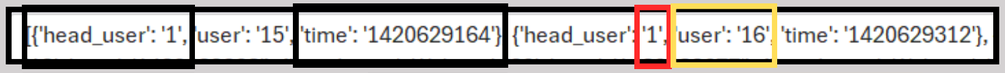


Social Heterogeneous Graph

1 ID / dataset

Propagation path

User ID

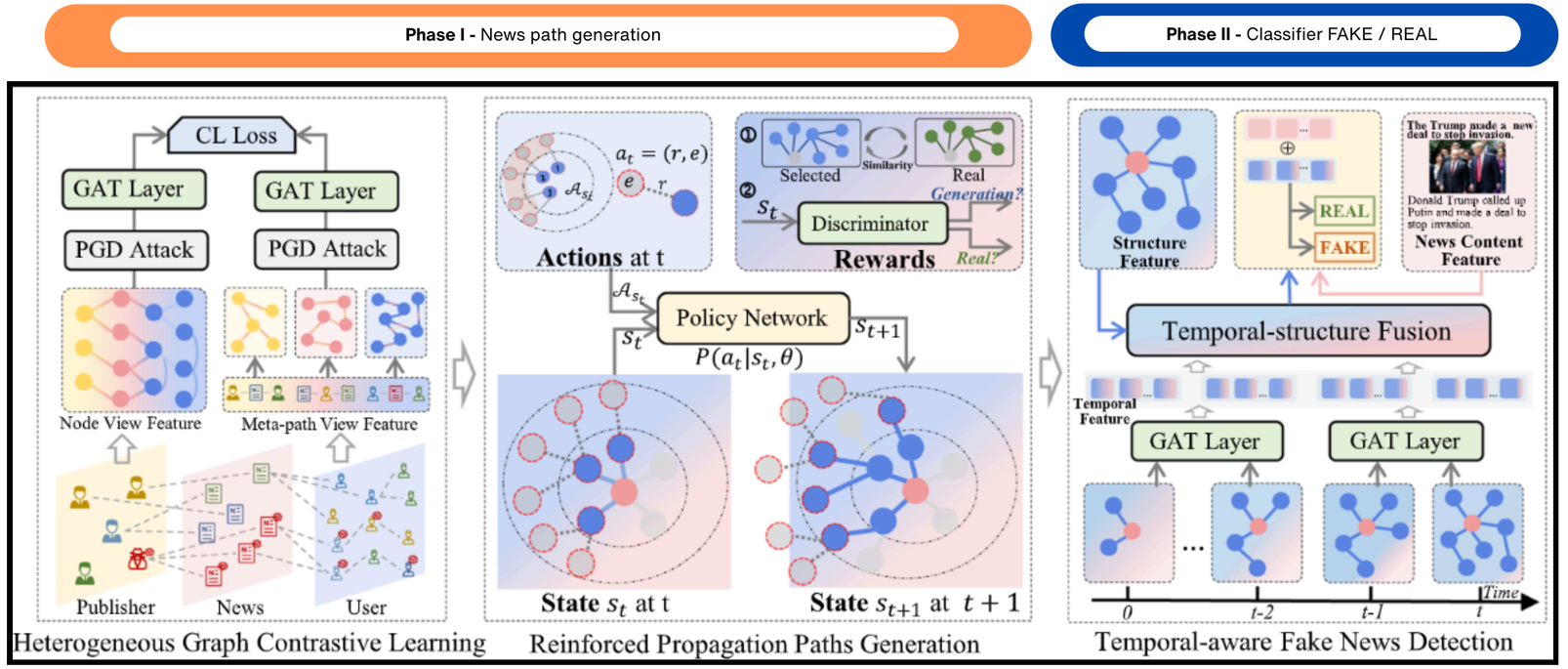


Head user

Time

Destinated user

Methodology



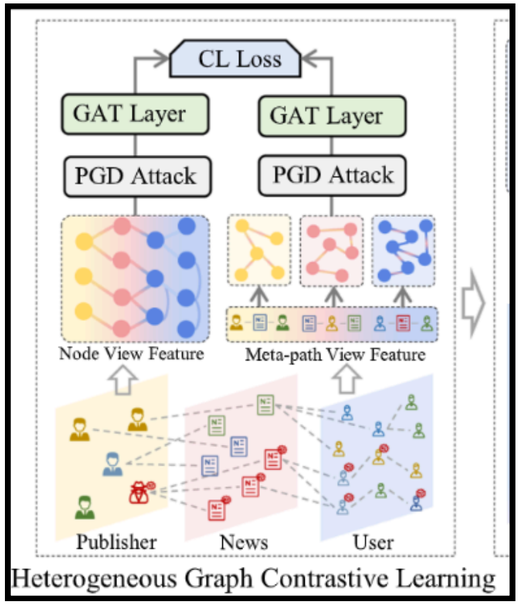
Phase I - News path generation

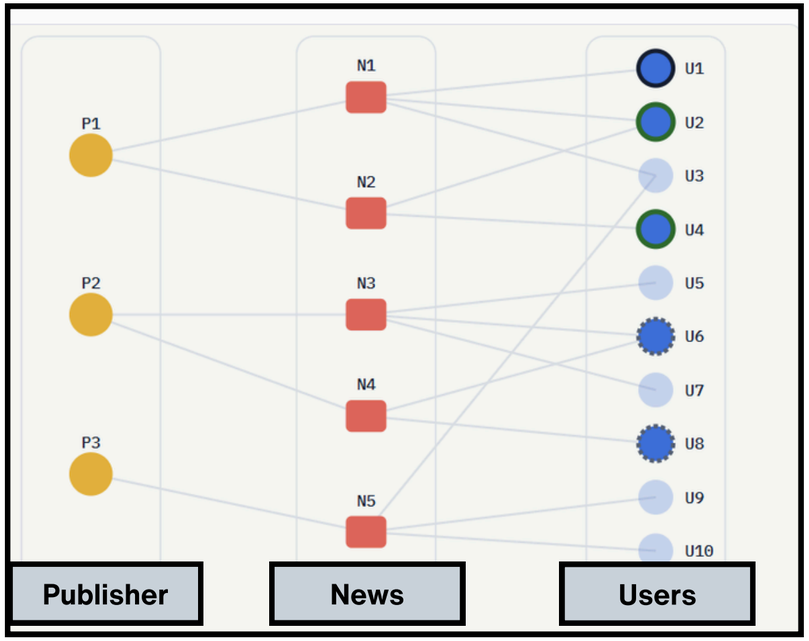
- Heterogeneous Graph Contrastive Learning
- Reinforced Propagation Paths Generations

Phase II - Classifier FAKE / REAL

- Temporal-aware Fake News Detection

Methodology





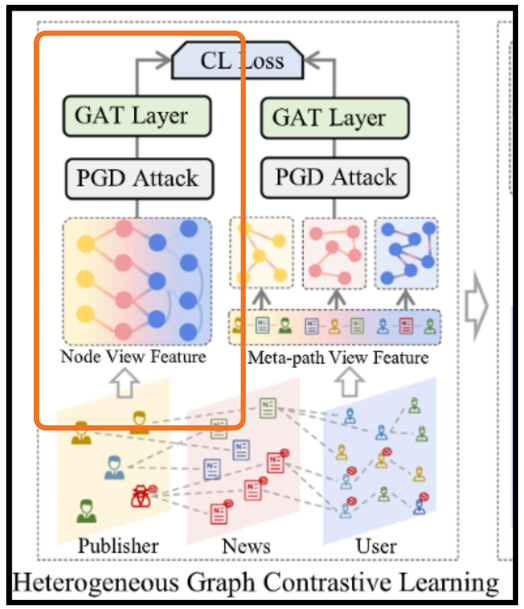
Phase I - News path generation - Heterogeneous Graph Contrastive Learning

Construct the Heterogeneous Graph with 3 node types—Publisher, News, and User—from platform-wide data, and **learn user embeddings without labels** through 2 views:

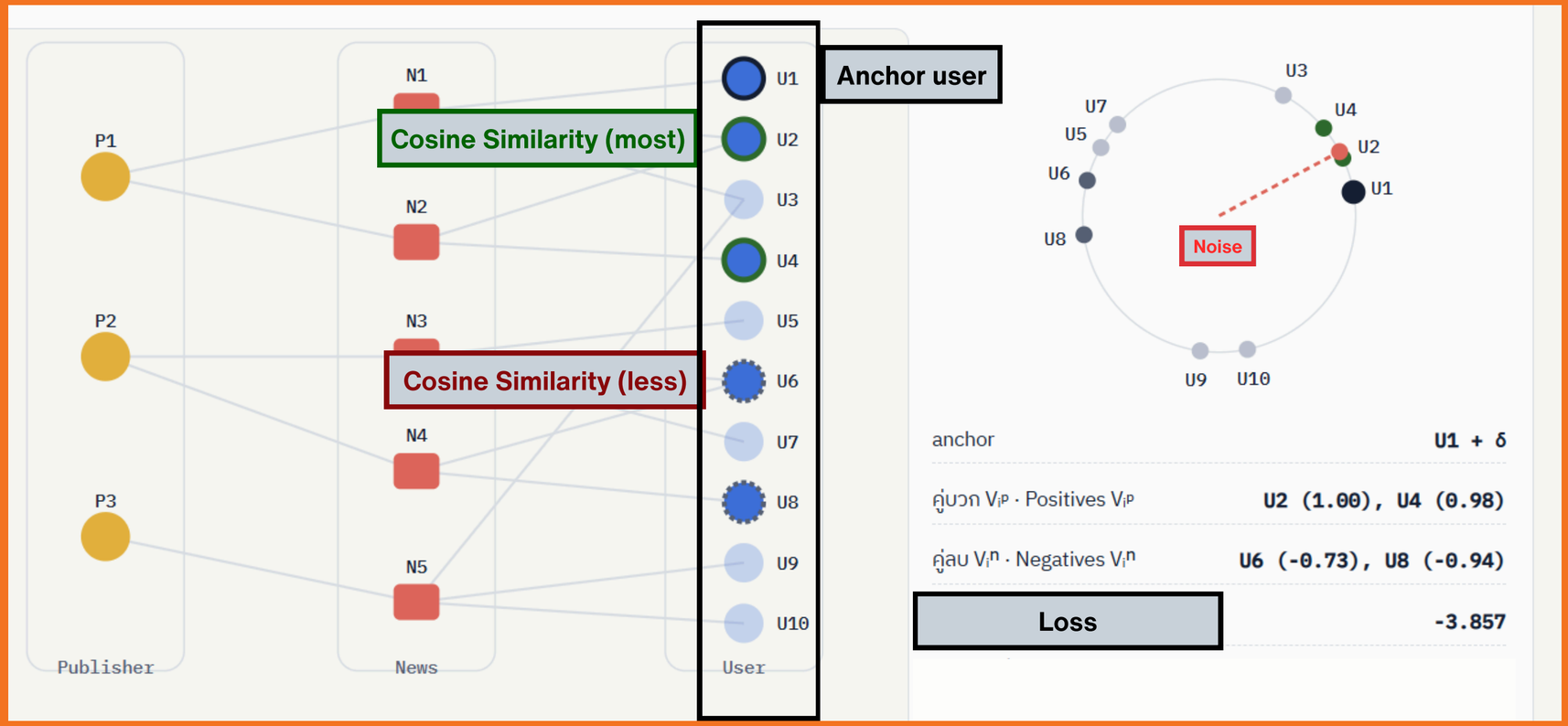
Node View Contrastive Learning

Meta-path View Contrastive Learning

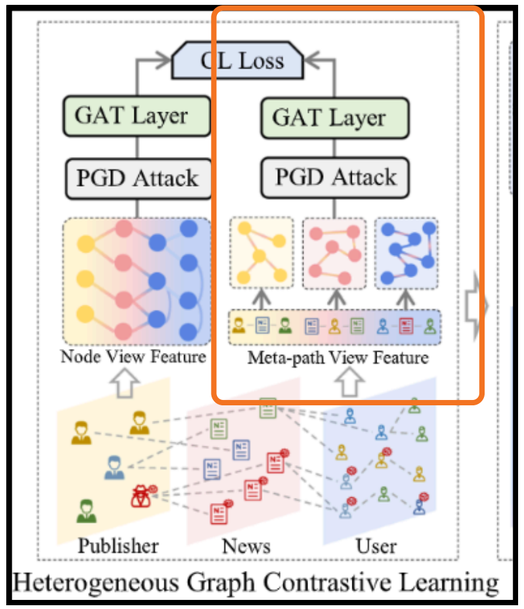
Methodology



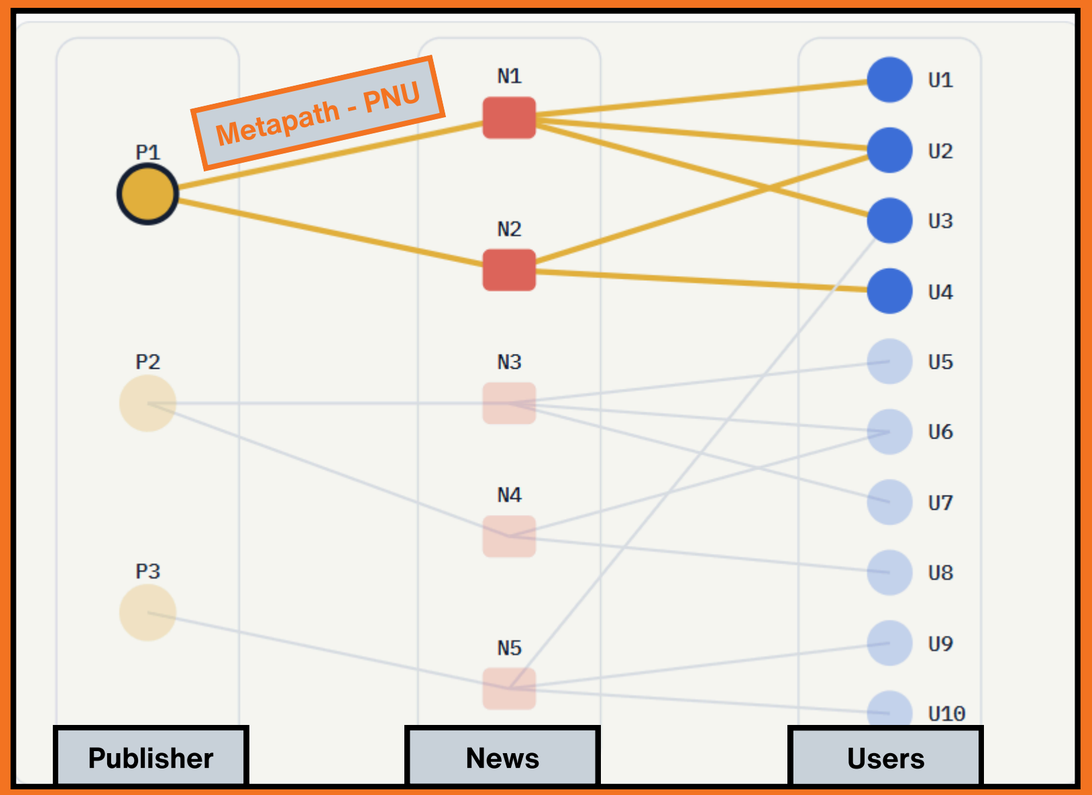
Node View Contrastive Learning



Methodology

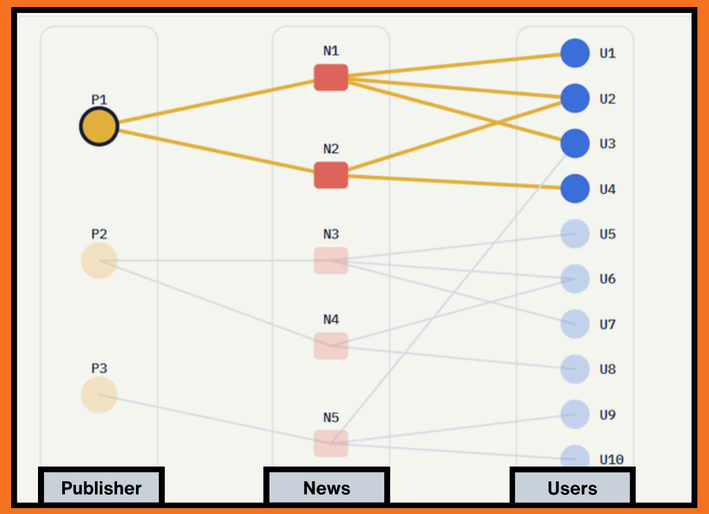


Meta-path View Contrastive Learning

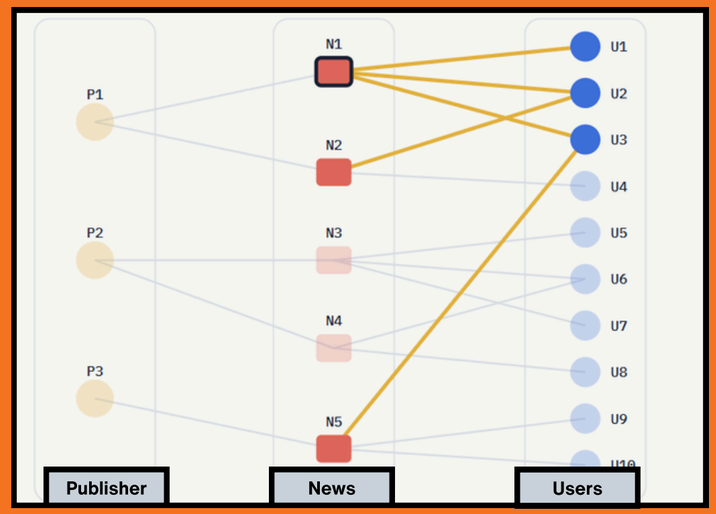


Methodology

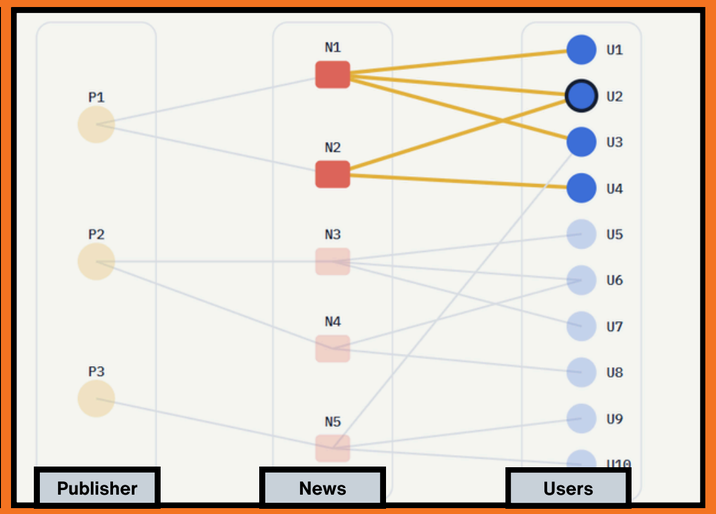
Meta-path View ContrastiveLearning



P → N → U

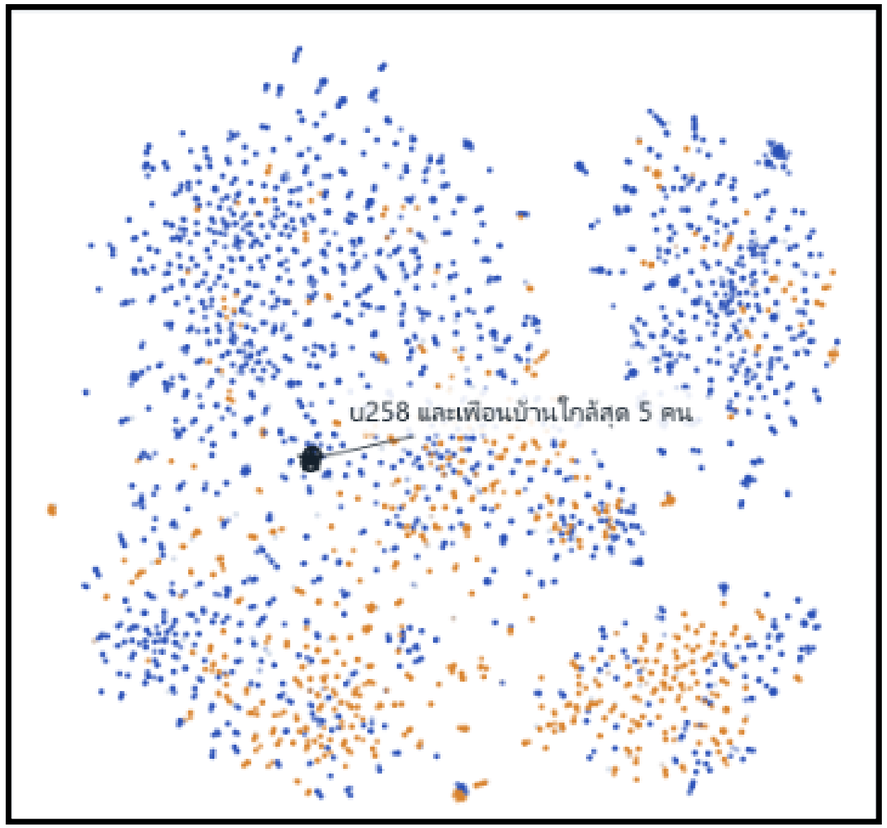


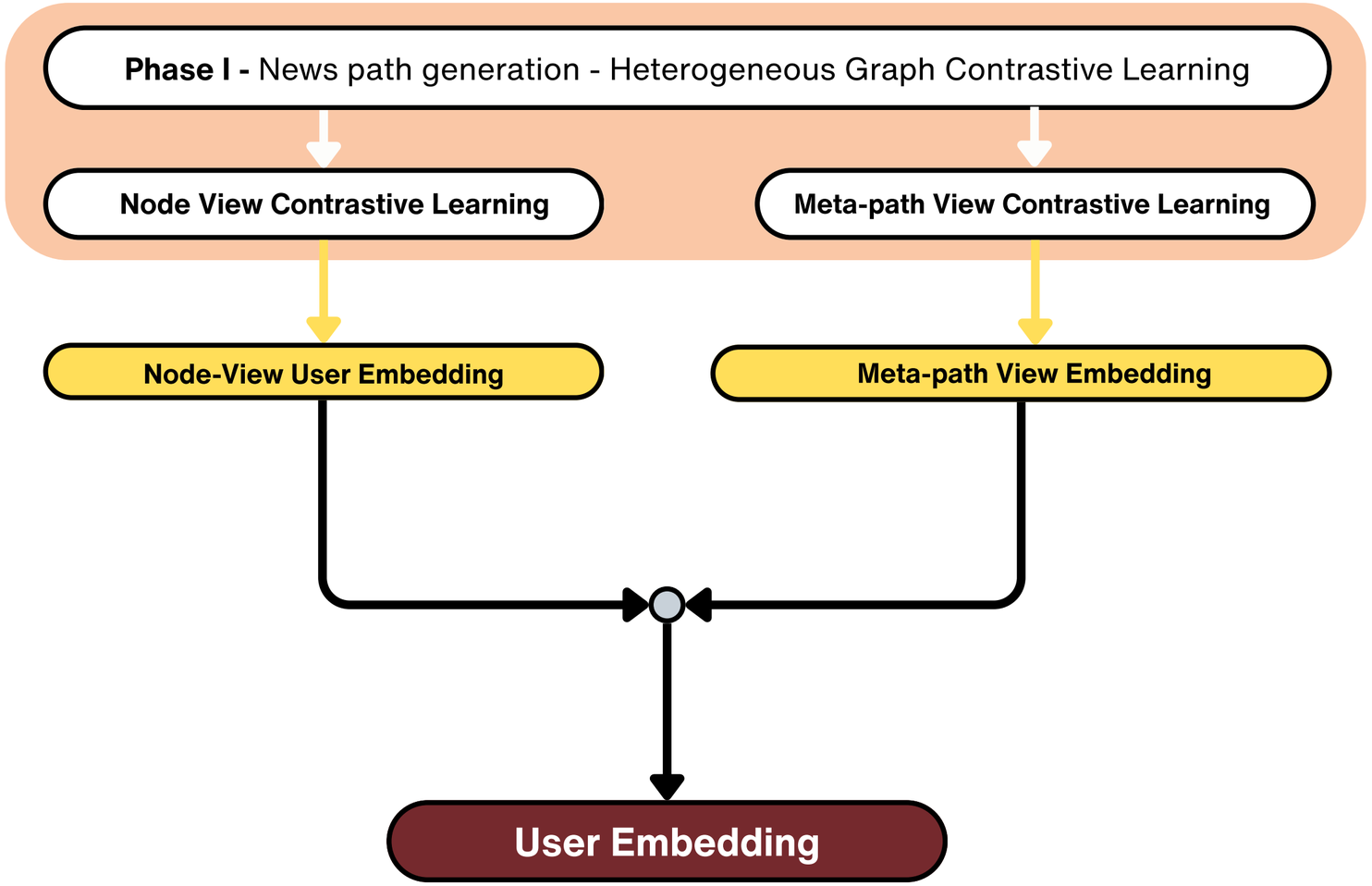
N → U → N



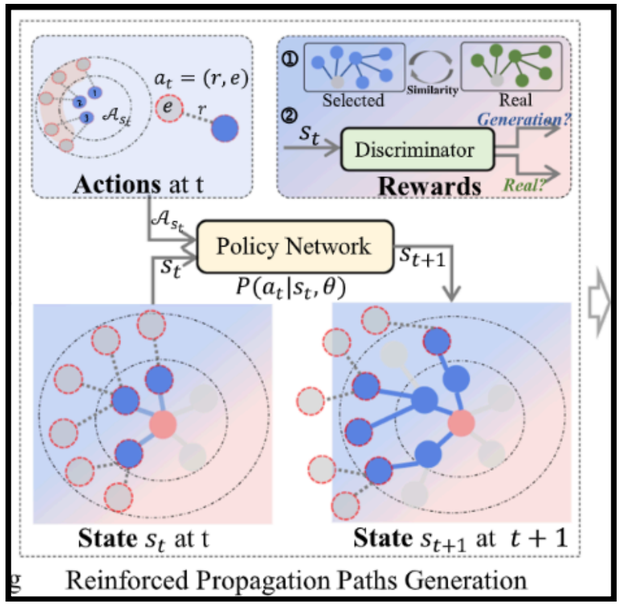
U → N → U

Methodology





Methodology



Phase I - News path generation - Reinforced Propagation Paths Generation

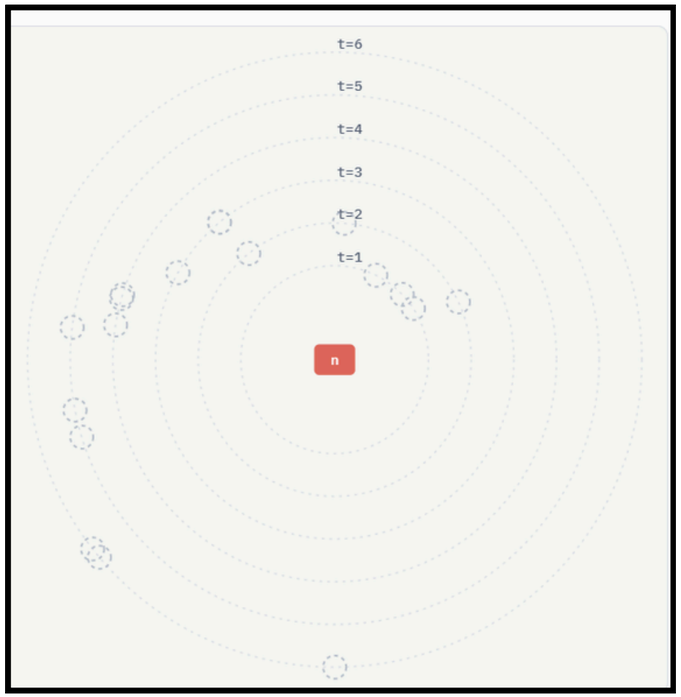
To simulate and generate **early-stage news propagation paths based on news content** without waiting for actual dissemination, thereby reducing social risks.

Policy Network

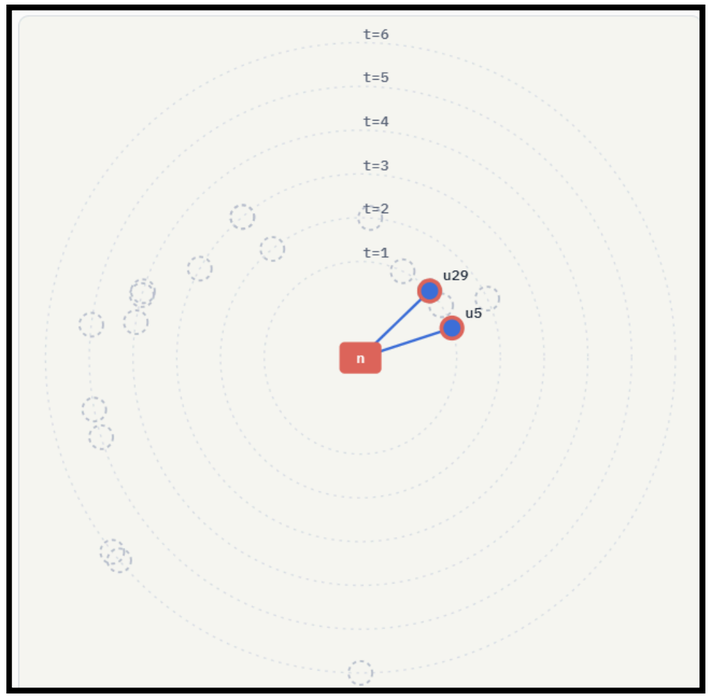
Methodology

Policy Network

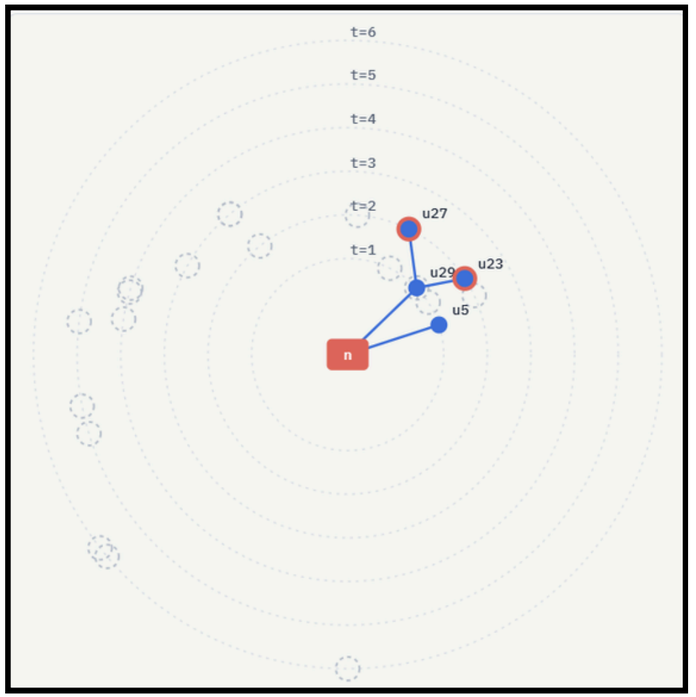
Time



t = 0



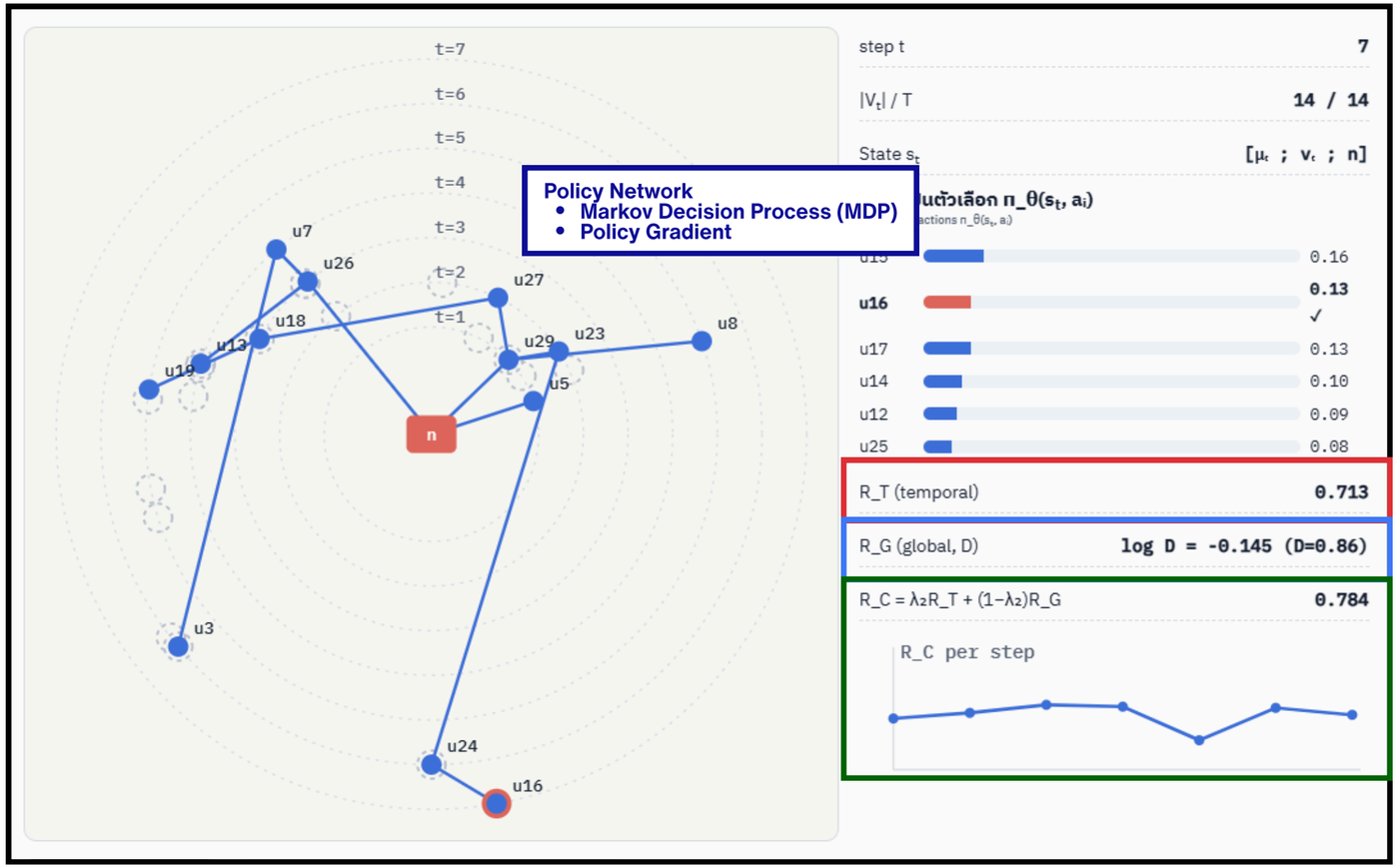
t = 1



t = 2

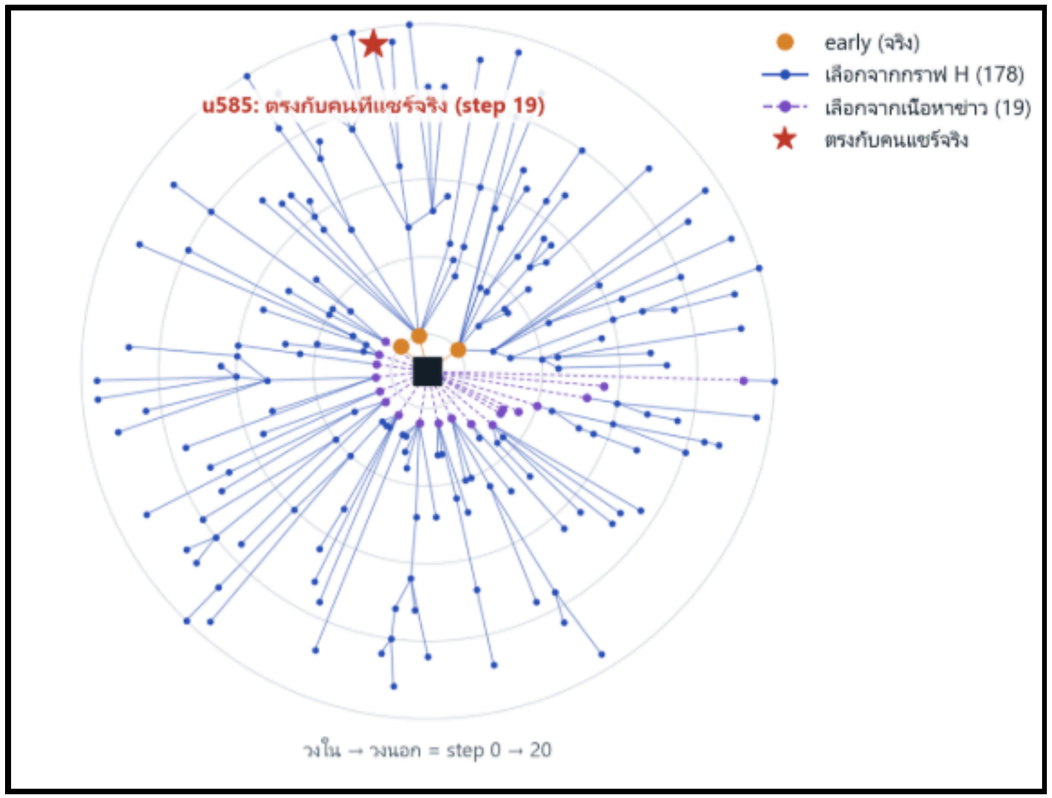
Methodology

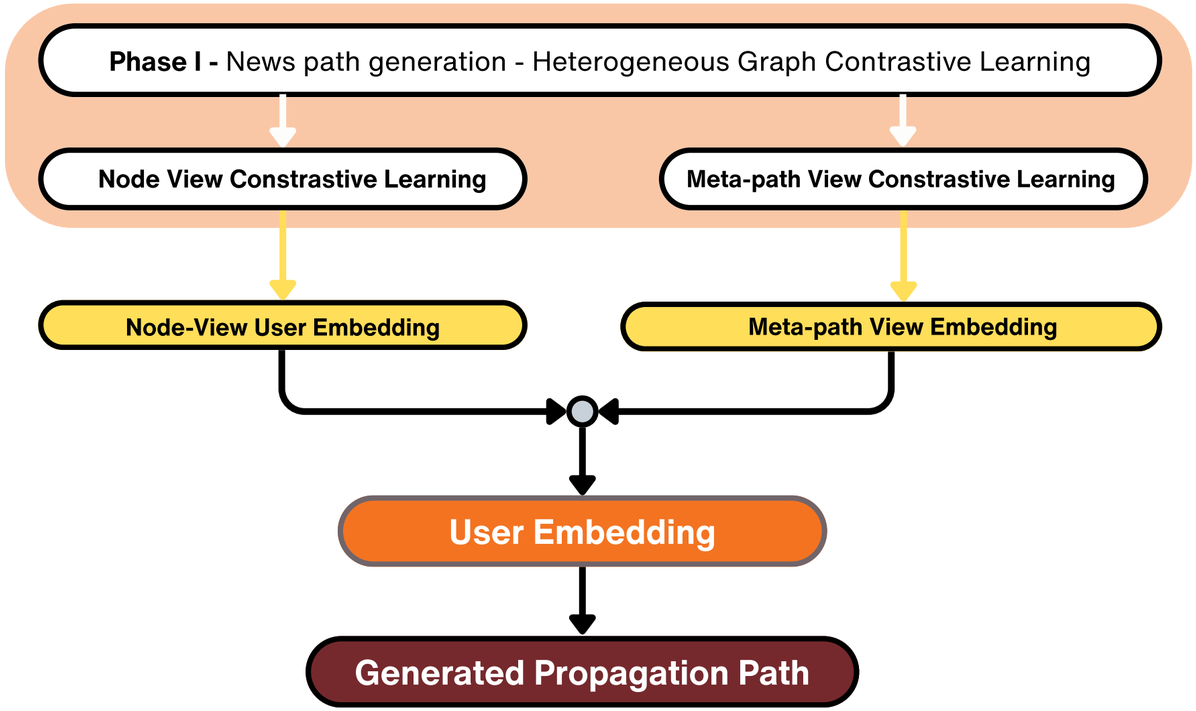
Phase I - News path generation - Reinforced Propagation Paths Generation



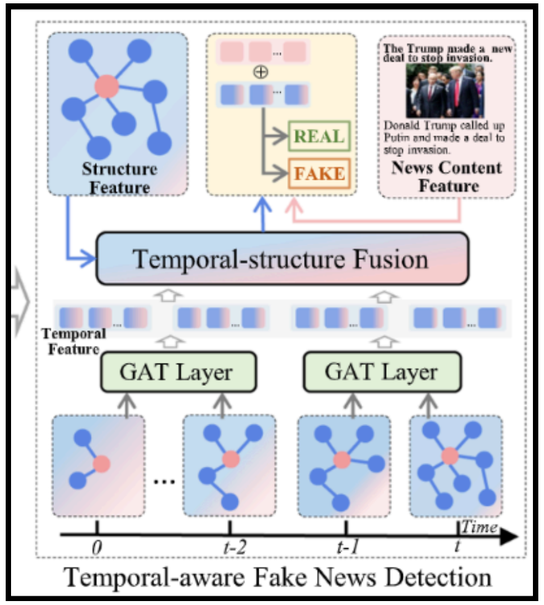
Methodology

Phase I - News path generation - Reinforced Propagation Paths Generation





Methodology



Phase II - Classifier - Temporal-aware Fake News Detection

Temporal-Aware Detection: **Dynamic Propagation Encoder**

Temporal-Aware Detection: Dynamic Propagation Encoder is a core sub-component of the RPPG-Fake model designed to **convert generated propagation paths into time-sensitive feature representations.**

Temporal-Aware Detection: **Fusion & Classification**

The final **classification component** of the RPPG-Fake framework.

Methodology

Temporal-Aware Detection: **Dynamic Propagation Encoder**

Self-Attention Mechanism

Early Stage

Time

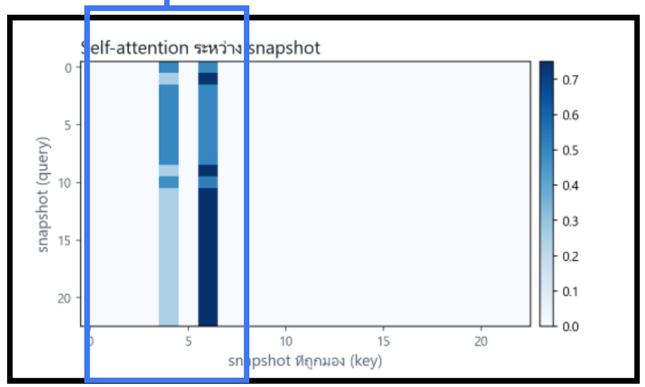
snapshot 0 · 2 node
early (จริง)

snapshot 2 · 4 node
early (จริง)

snapshot 7 · 54 node
+ G'

snapshot 14 · 124 node
+ G'

snapshot 22 · 201 node
+ G'



Predicted Target (Key) from the most attentions.

Generated User

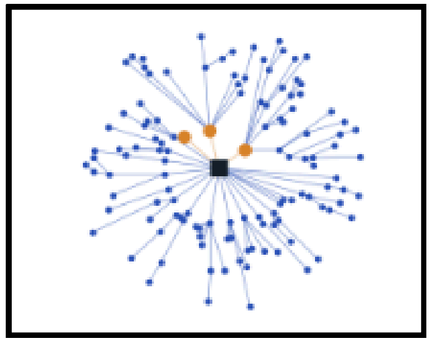
Real user

News

Methodology

Temporal-Aware Detection: **Fusion & Classification**

Fusion & Classification



A single representative vector of the overall propagation

+

Visual content

BREAKING: Pope Francis Just Backed Trump, Released Incredible Statement Why- SPREAD THIS EVERYWHERE

Textual content

Trump is getting support from every leader, and that's the support that will make him grow great and strong. These elections will bring an immense change in our country.

$$P(y \mid n, G)$$

Approch to 1: Predict as the **REAL** news
Approch to 0: Predict as the **FAKE** news

Cross-Entropy Loss

This loss value will be used to update the weights of the entire model.

Data

Model training

News

BREAKING: Pope Francis Just Backed Trump, Released Incredible Statement Why- SPREAD THIS EVERYWHERE

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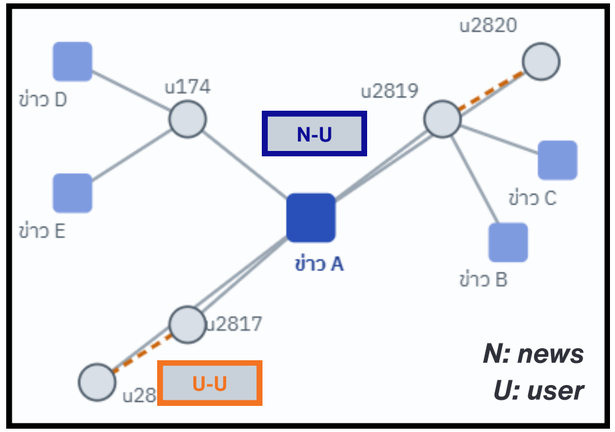
Visual content

Textual content

1 H graph / dataset

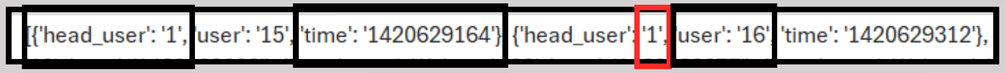


Social Heterogeneous Graph

1 ID / dataset

Propagation path

User ID



Head user | **Time** | **Destinated user**

Data

Final Model

News

BREAKING: Pope Francis Just Backed Trump, Released Incredible Statement Why- SPREAD THIS EVERYWHERE

Trump is getting support from every leader, and that's the support that will make him grow great and strong. These elections will bring an immense change in our country.

Visual content

Textual content

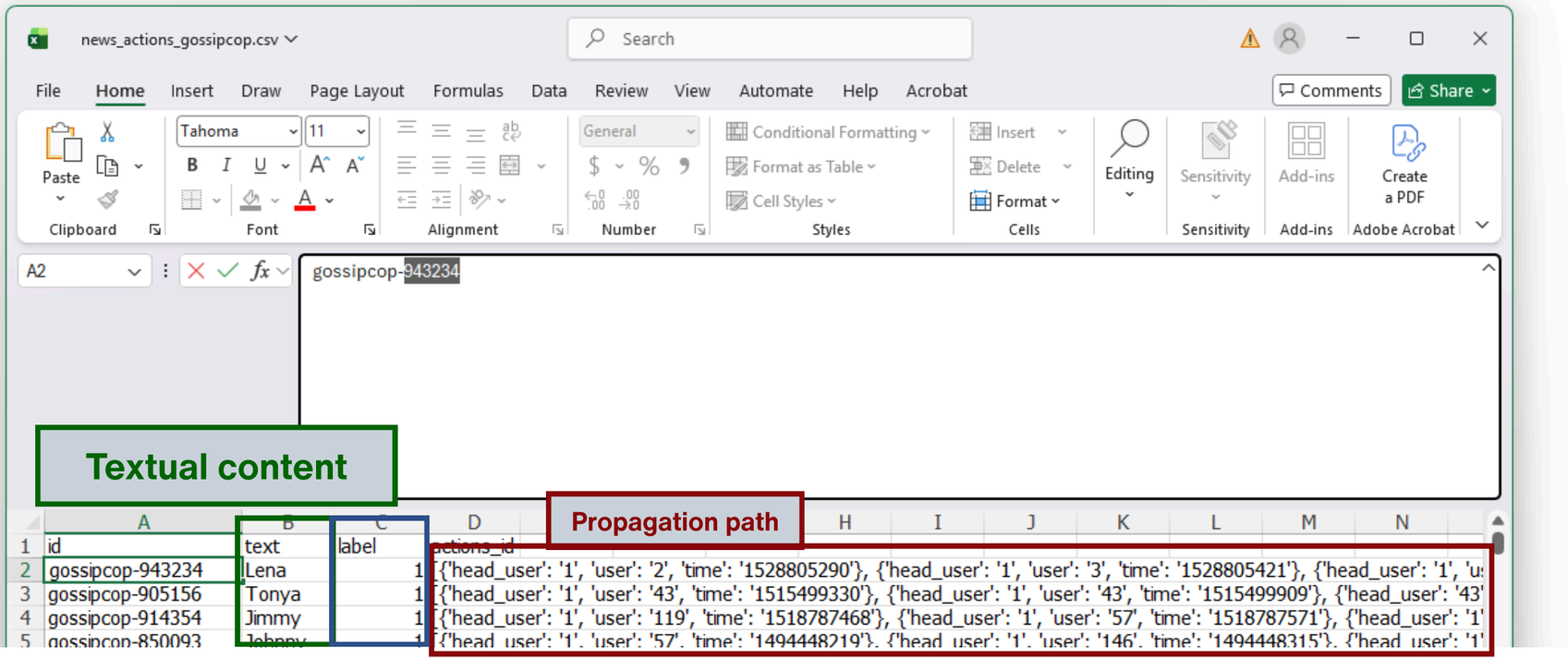
Early Propagation Path *(3-5 shared)*

index	id	label	text
1937	5.44E+17	fake	Aussie grand mufti & Imam Council condemn #Sydneysiege hostage capture http://t.co/m51P8dUPhB - LIVE UPDATES http://t.co/tTw3kMkYTZ
1921	5.44E+17	fake	Two female hostages have run out of #SydneySiege. 5 in total have escaped http://t.co/isvMtY4oUQ http://t.co/Xv2gJ1Q3zZ
1991	5.45E+17	fake	Hostages seen fleeing from cafe at centre of #SydneySiege moments ago http://t.co/nM2pss0I9d http://t.co/MXGOxDGdwv

Input (Train)

gossipcop-943234.jpg

Visual content



y: News Label ; 1 = Real, 0 = Fake

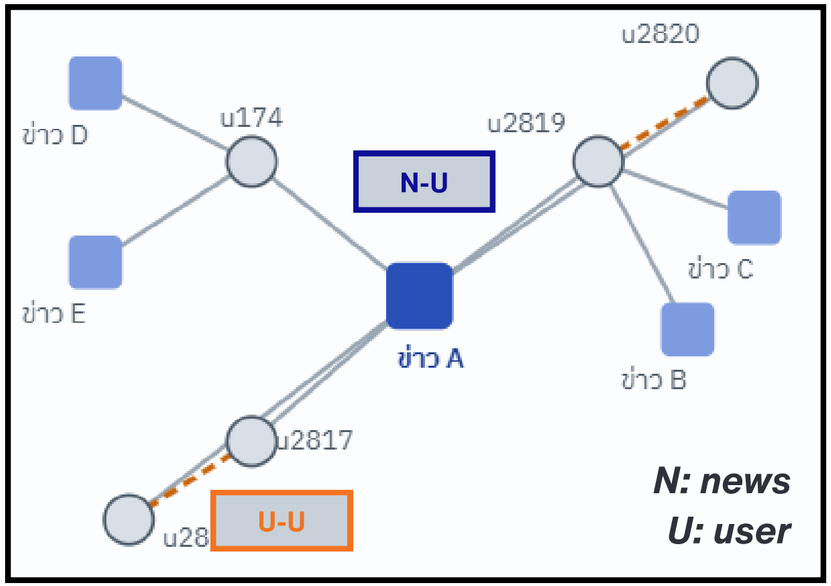
Data

Input (Train)

build_graph(), *a command.*

```
# src/data/components/politifact.py → build_graph()
NEWS_ROOT = 1                                   #
head_user = 1 หมายถึงตัวข่าวเอง
src, dst, rel = [], [], []
for i in train_idx:                             # ข่าว
train เท่านั้น
    news_node = num_users + i                   # node
ของข่าว = ต่อท้าย user
    users = set()
    for head, user, time in news[i]["eng"]:
        users.add(user)
        if head != NEWS_ROOT:                   # แชร์ต่อ
จาก user อื่น
            src.append(head); dst.append(user);
rel.append(0)   # 0 = repost (U-U)
    for user in users:
        src.append(news_node); dst.append(user);
rel.append(1)  # 1 = แชร์ข่าว (N-U)

edge_index = [src + dst, dst + src]             # เติม
เส้นย้อนทิศ
edge_type  = rel + rel
```



Social Heterogeneous Graph

Experiment

Dataset	Pheme	PolitiFact	GossipCop
#. total news	3,506	478	5,526
#. fake news	1,142	294	1,935
#. real news	2,364	184	3,591
#. words per news	13.8	1545.8	578.6

Pheme	This dataset is based from the **Twitter**, a widely used social media platform.
Politifact	This dataset is collected from **fact-check websites.**
GossipCop	This dataset is collected from **fact-check websites.**

Experiment

Model baseline

Content-based Detection Model

- **BERT** (2018)
- **MVAE** (2019)
- **EANN** (2018)
- **CAFE** (2022)
- **LLAMA2** (2022)
- **Detect-GPT** (2022)

Propagation Enhanced Detection Model

- **CSI** (2017)
- **Bi-GCN** (2020)
- **UPFD** (2021)
- **MFAN** (2022)

Experiment

Implementation Details

All dataset are randomly split into training, testing, and validation set which following ratio.

Holdout Split (Random Split)

Train (7)	**Test** (2)	**Val.** (1)
For updating the **weight** and parameters in the model while training.	For testing the model performance using the unseen data	For tuning the performance while model is training for adjust the hyperparameters.

Experiment

Implementation Details

All dataset are randomly split into training, testing, and validation set which following ratio.

Holdout Split (Random Split)

Validation (1)

- **AdamW** as the optimizer
  - The batch size: 16
  - The initial learning rate: $5e^{-3}$
  - The hidden size of user embedding: 128

Implement based on

PyTorch 1.13.0 on
Linux server equiped with
GeForce RTX3090 GPUs

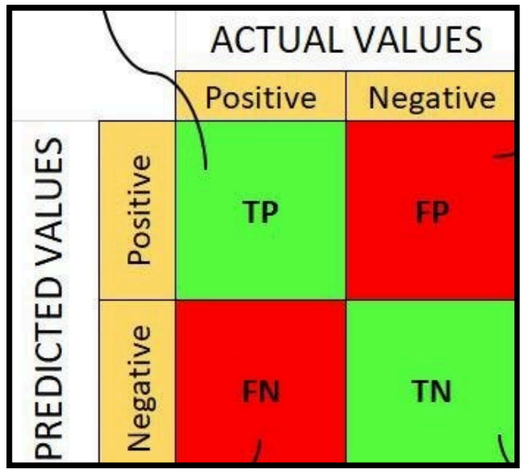
Result

Performance with **Full propagation path**

TABLE II
PERFORMANCE WITH FULL PROPAGATION PATHS

Category	Method	PolitiFact				GossipCop				Pheme			
		Acc	Pre	Rec	F1	Acc	Pre	Rec	F1	Acc	Pre	Rec	F1
Propagation Enhanced	CSI	0.753	0.834	0.742	0.758	0.782	0.823	0.783	0.808	0.771	0.825	0.831	0.828
	Bi-GCN	0.788	0.834	0.742	0.758	0.890	**0.928**	0.848	0.890	0.827	0.840	0.834	0.835
	UPFD	0.904	0.923	0.857	0.889	0.879	0.906	0.931	0.918	–	–	–	–
	MFAN	0.872	0.841	0.881	0.860	0.862	0.903	0.910	0.906	0.823	0.878	0.847	0.862
Proposed	**RPPG-Fake**	**0.924**	**0.925**	**0.902**	**0.914**	**0.891**	0.911	**0.940**	**0.925**	**0.844**	**0.891**	**0.868**	**0.879**

Best results are in bold and second best results are underlined.



Accuracy

measures the percentage of **correct predictions out of total predictions.**

$$\frac{TP + TN}{TP + TN + FP + FN}$$

Recall

measures the ability of a **classification model to find all actual positive instances in a dataset**

$$\frac{TP}{TP + FN}$$

Precision

measures the exactness of a machine learning **model's positive predictions**

$$\frac{TP}{TP + FP}$$

F1 Score

measures a classification model's accuracy by **combining precision and recall** into a single harmonic mean

$$2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

Result

Performance with **Fewer Propagation Paths at early stage**

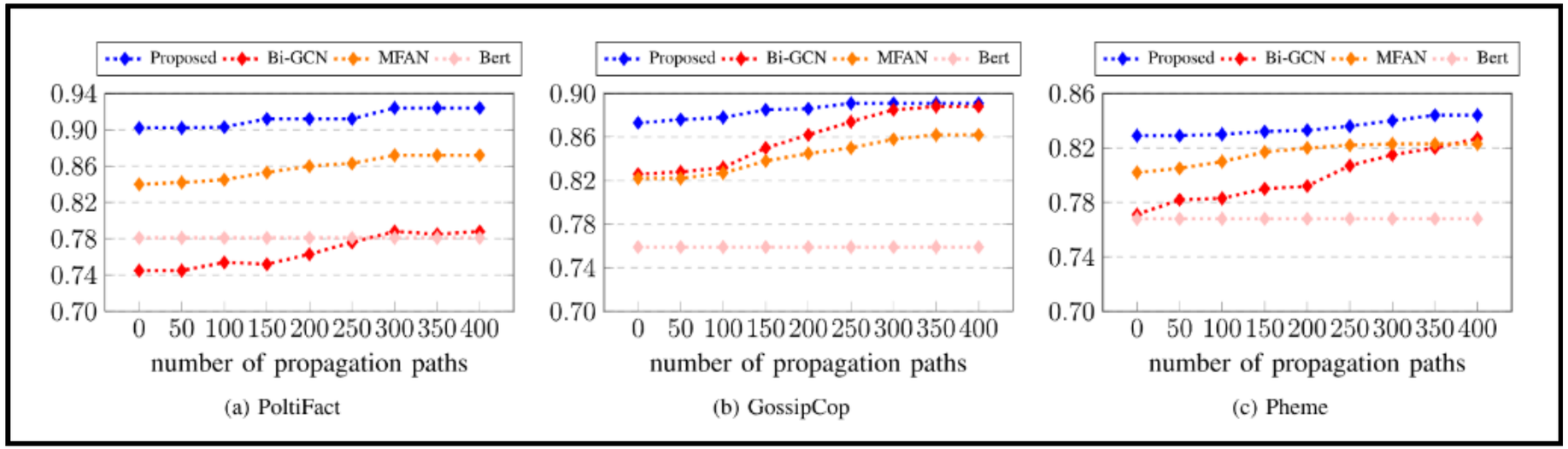
TABLE III
PERFORMANCE WITH FEWER PROPAGATION PATHS AT EARLY STAGE

Category	Method	PolitiFact				GossipCop				Pheme			
		Acc	Pre	Rec	F1	Acc	Pre	Rec	F1	Acc	Pre	Rec	F1
Content Based	BERT	0.781	0.766	0.878	0.818	0.759	0.787	0.764	0.775	0.768	0.704	0.778	0.739
	MVAE	0.812	0.803	0.835	0.819	0.782	0.802	0.751	0.776	0.776	0.734	0.722	0.728
	EANN	0.804	0.808	0.794	0.781	0.796	0.812	0.765	0.788	0.771	0.714	0.701	0.704
	CAFE	0.848	0.857	0.850	0.853	0.832	0.804	0.903	0.851	–	–	–	–
	LLAMA2	0.848	0.865	0.780	0.820	0.825	0.811	0.872	0.840	–	–	–	–
	Detect-GPT	0.858	0.888	0.780	0.831	0.831	0.833	0.884	0.858	–	–	–	–
Propagation Enhanced	CSI	0.734	0.774	0.571	0.658	0.753	0.748	0.764	0.756	0.754	0.799	0.839	0.819
	Bi-GCN	0.745	0.696	0.762	0.727	0.826	0.871	0.881	0.876	0.771	0.825	0.831	0.828
	UPFD	0.862	0.892	0.786	0.835	0.822	0.860	0.891	0.875	–	–	–	–
	MFAN	0.840	0.846	0.787	0.815	0.822	0.861	0.888	0.875	0.802	0.853	0.846	0.850
Proposed	**RPPG-Fake**	**0.902**	**0.921**	**0.853**	**0.886**	**0.873**	**0.904**	**0.921**	**0.913**	**0.829**	**0.881**	**0.854**	**0.867**
	Imp.(%)	*+4.0*	*+2.9*	*+0.3*	*+3.3*	*+4.1*	*+3.3*	*+1.8*	*+3.7*	*+2.7*	*+2.8*	*+0.9*	*+1.7*

Best results are in bold and second best results are underlined.

Result

The accuracy performance with different generation users



Ablation Study

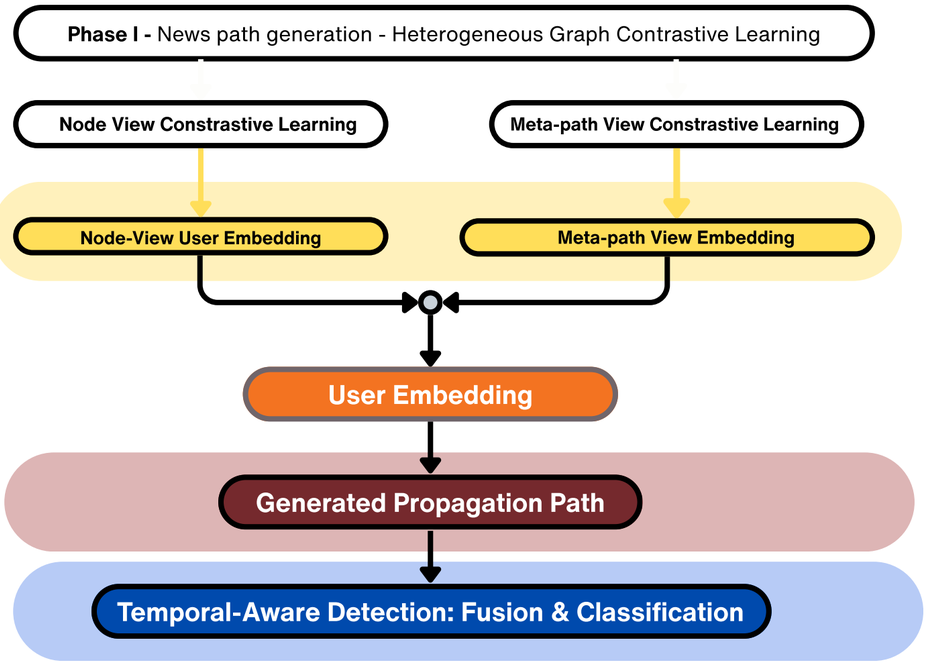


TABLE IV
ABLATION STUDIES OF *RPPG-FAKE* ON THREE DATASETS

Abaltion Categoty	Method	PolitiFact				GoosipCop				Pheme			
		Acc	Pre	Rec	F1	Acc	Pre	Rec	F1	Acc	Pre	Rec	F1
Proposed	RPPG-Fake	**0.924**	**0.925**	**0.902**	**0.914**	**0.891**	**0.911**	**0.940**	**0.925**	**0.844**	**0.891**	**0.868**	**0.879**
Heterogeneous Contrastive Learning	w/o all CL	0.891	0.919	0.829	0.872	0.874	0.910	0.921	0.915	0.827	0.879	0.852	0.867
	w/o Node-view CL	0.902	0.921	0.854	0.886	0.876	0.905	0.924	0.914	0.828	0.881	0.852	0.867
	w/o Meta-path CL	0.913	0.902	0.902	0.902	0.885	0.907	0.936	0.921	0.831	0.883	0.856	0.869
Propagation Path Generation	Random Sample	0.891	0.897	0.854	0.875	0.872	0.907	0.917	0.912	0.824	0.879	0.849	0.864
	Node Similarity	0.902	0.921	0.854	0.886	0.873	0.905	0.921	0.913	0.826	0.881	0.850	0.865
Temporal-aware Detection	w/o structure feature	0.913	0.923	0.878	0.900	0.886	0.908	0.937	0.922	0.834	0.885	0.858	0.871
	w/o temporal feature	0.902	0.900	0.878	0.889	0.884	0.906	0.935	0.920	0.836	0.887	0.859	0.873

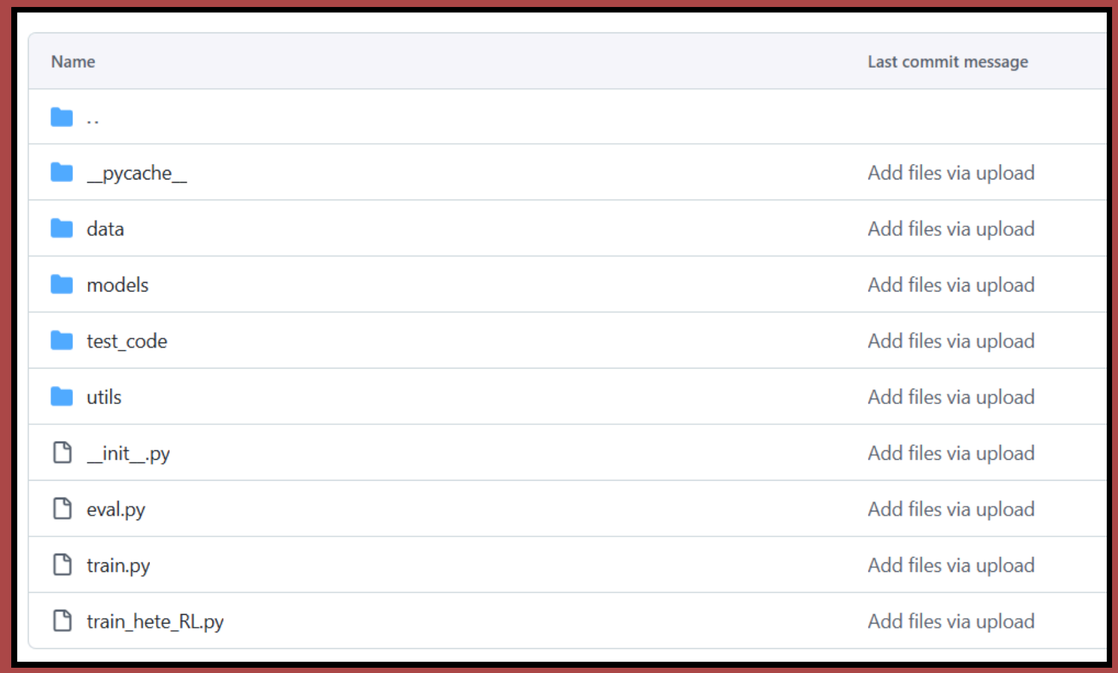
Conclusion

- **Core Framework**
  - Introduces RPPG-Fake, a framework that simulates news propagation paths using **content features** and integrates them into a **temporal-aware detection module.**
- **Evaluation**
  - Validated across three popular benchmark datasets, demonstrating significantly improved detection performance.
- **Future Directions**
  - Suggests enhancing path generation with richer social dynamics, incorporating adaptive learning for better cross-domain generalization, and optimizing real-time performance and scalability for large-scale social networks.

Limitation & Critique

Supplementary Code

The Configuration files and some essential source code components from official GitHub are missing such as lacks code implementation for the Reward and Discriminator modules.

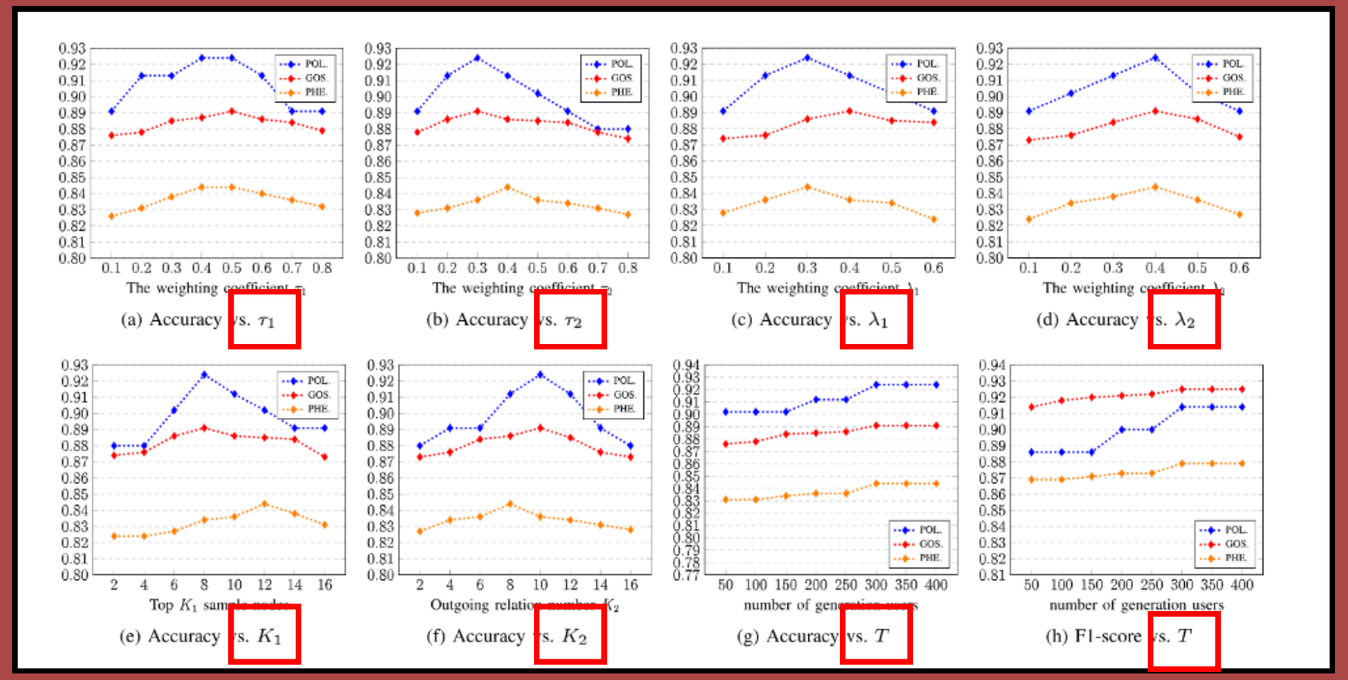


Name	Last commit message
..	
__pycache__	Add files via upload
data	Add files via upload
models	Add files via upload
test_code	Add files via upload
utils	Add files via upload
__init__.py	Add files via upload
eval.py	Add files via upload
train.py	Add files via upload
train_hete_RL.py	Add files via upload

Limitation & Critique

Parameter Setting

Some hyperparameter are only ploted without explicit numerical data. Including, key parameters are omitted, including RL epochs, dropout rates, GAT attention heads, and PGD values



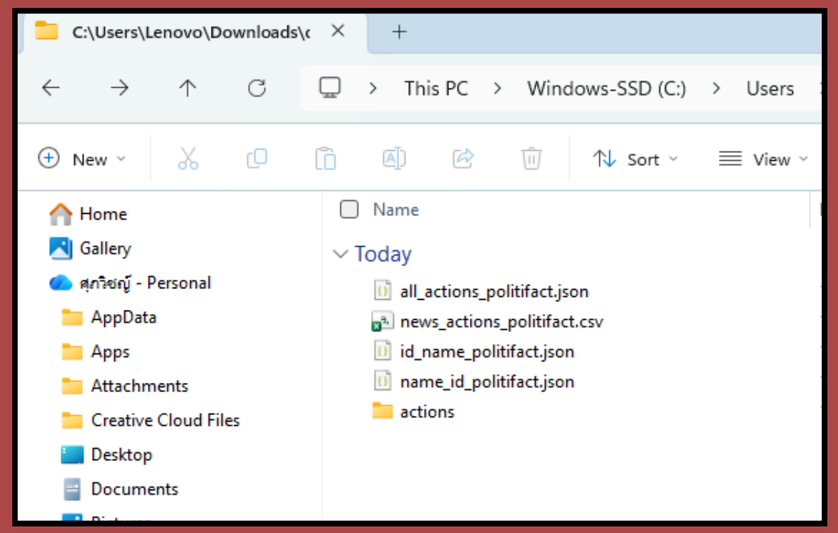
(a) Accuracy vs. τ_1 (b) Accuracy vs. τ_2 (c) Accuracy vs. λ_1 (d) Accuracy vs. λ_2

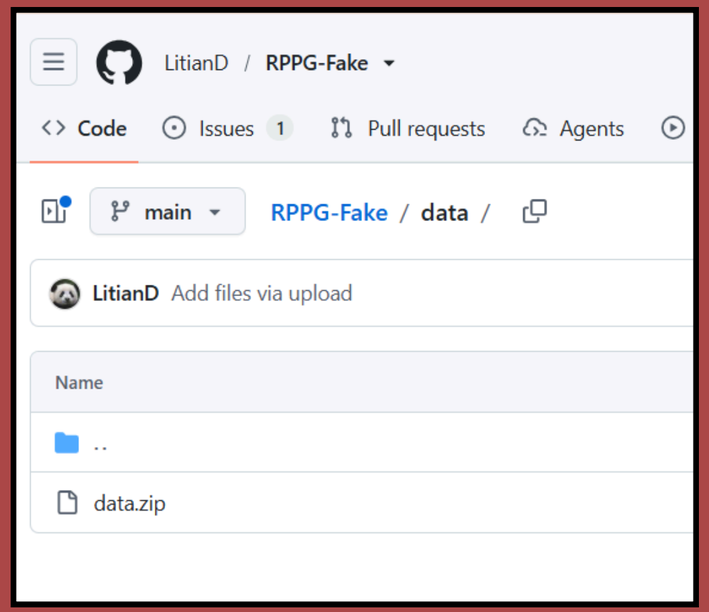
(e) Accuracy vs. K_1 (f) Accuracy vs. K_2 (g) Accuracy vs. T (h) F1-score vs. T

Limitation & Critique

Incomplete Dataset

The provided repository contains only 483 news samples for PolitiFact. Including, lacks publisher information (all cascades share the same initial user node), making it impossible to construct Publisher-News-User (PNU) meta-paths. And GossipCop and PHEME datasets are not provided in the repository

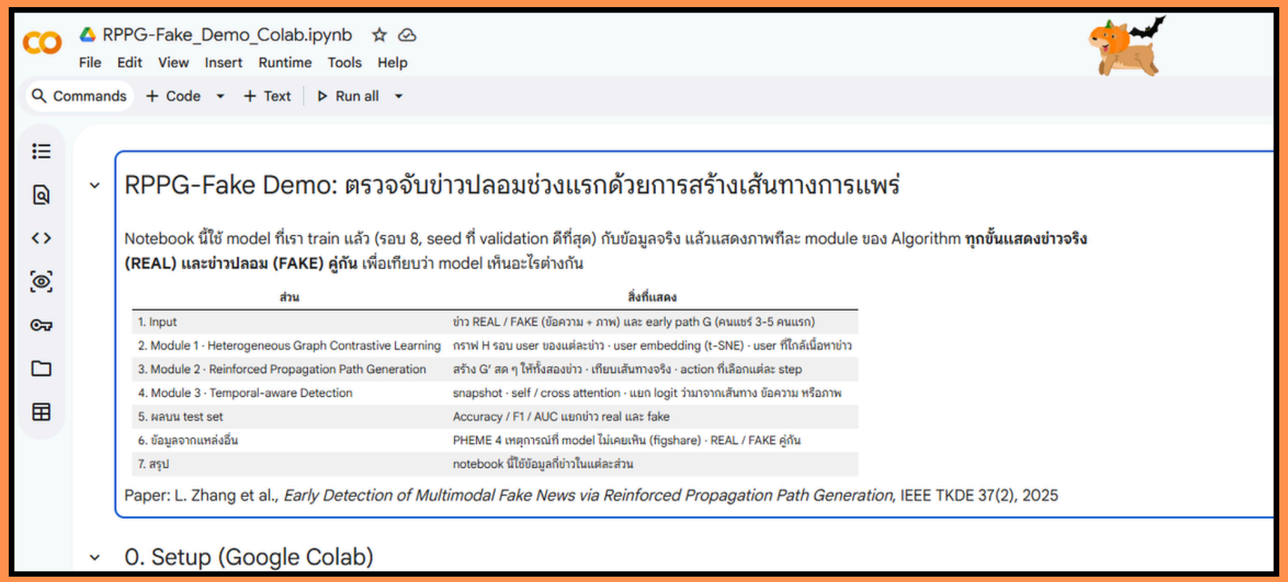




Limitation & Critique

Demo Limitation

From this demo about the code, the PC resource doesn't have much more. Therefore, some train runs will not based on the paper such as this demo using only 3 seed runs, 5 epochs.



Limitation & Critique

หัวข้อ	Paper	ของเรา (รอบ 8)
PHEME	3,506 ข่าว	2,004 ข่าว · เฉพาะที่มีครบ ข้อความ + ภาพ + เส้นทาง
GossipCop	5,526 ข่าว · real 65%	2,827 ข่าว · real 51% · ดึงเนื้อข่าวและภาพเอง
PolitiFact	478 ข่าว · มีภาพ	483 ข่าวของผู้เขียน · ไม่มีภาพ
Meta-path	PNU · NUN · UNU	UU · NUN · UNU (ข้อมูลไม่มี publisher)
Encoder	ไม่ได้เปิด code / weights	CLIP ViT-B/32 ทั้งข้อความและภาพ
จำนวนรอบ	5 รอบ · รายงานค่าเฉลี่ย	3 seeds · สุ่ม split ใหม่ทุก seed · รายงาน ± std
Module 1 / 2	ไม่ระบุ step / epoch	contrastive 300 step · RL 5 epoch (รอบ 4 ใช้ 10)
Module 3	ไม่ระบุ epoch	≤ 30 epoch · early stop patience 10 · เลือกด้วย val acc
τ1 τ2 λ1 λ2 K1 K2 T	มีแค่กราฟ Fig. 4	0.4 · 0.3 · 0.4 · 0.4 · 8 · 10 · 200 (จุดสูงสุดของ Fig. 4)
แก้เพิ่ม	—	z-score ก่อน Eq. 7 · node view ใช้โครงสร้าง · user ใหม่ = UNK
เครื่อง	RTX 3090 · PyTorch 1.13	RTX 5050 Laptop 8 GB · RAM 16 GB · PyTorch 2.11
เหมือนกัน	AdamW lr 5e-3 · batch 16 · hidden 128	split 7:1:2 · early stage = 3–5 engagements แรก

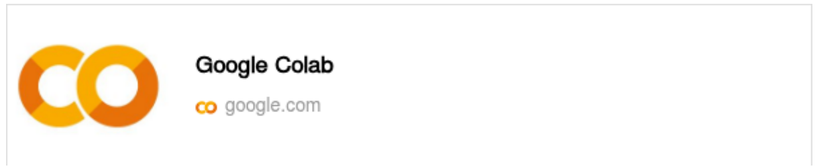
✕ Incomplete Dataset

✕ Parameter Setting

Limitation & Critique

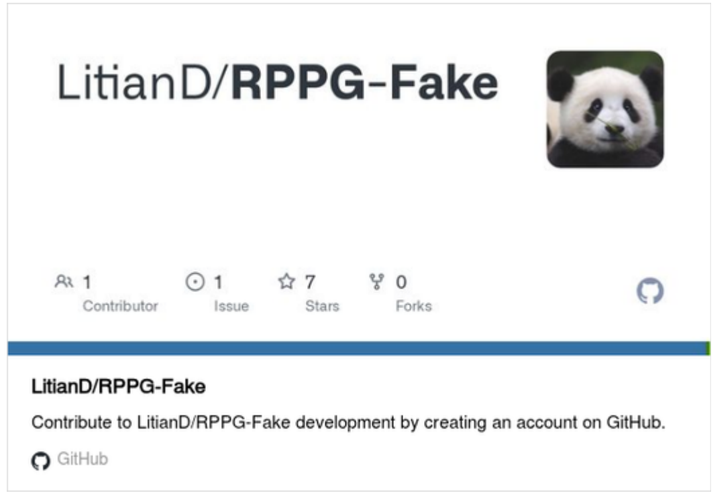
Dataset	early Acc	paper	Δ pp	early F1	paper	full Acc	paper	Δ pp	ไม่ใช้ G′
PolitiFact	0.826 ±0.04	0.902	−7.6	0.781	0.886	0.868	0.924	−5.6	0.868
GossipCop	0.878 ±0.09	0.873	+0.5	0.879	0.913	0.908	0.891	+1.7	0.900
PHEME	0.837 ±0.01	0.829	+0.8	0.886	0.867	0.844	0.844	0.0	0.844

Source Code



Google Colab

GitHub



Paper’s GitHub

THANK YOU

FINAL
PRESENTATION

EARLY DETECTION OF MULTIMODAL FAKE NEWS

VIA REINFORCED PROPAGATION PATH GENERATION

L. ZHANG ET AL. · IEEE TKDE 37(2), 2025

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